

触摸测试 (evdev)

命令行测试

确定输入节点

```
ls /dev/input/  
cat /proc/bus/input/devices
```

查看原始触摸数据

```
hexdump /dev/input/event4
```

编码测试

touch_test.c (lvgl_100ask_course_v9\part3\02_codes\lesson_2-3\touch_test.c)

```
#include <stdio.h>  
#include <stdlib.h>  
#include <fcntl.h>  
#include <unistd.h>  
#include <unistd.h>  
#include <linux/input.h>  
  
/*  
# 编译  
export PATH=$PATH:/home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-  
V1/prebuilt/rootfsbuilt/arm/toolchain-sunxi-glibc-gcc-830/toolchain/bin  
export CROSS_COMPILE=arm-openwrt-linux-gnueabi  
export STAGING_DIR=/home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-  
V1/out/t113_s4/100ask_lvgl9/openwrt/staging_dir/target  
  
arm-openwrt-linux-gnueabi-gcc touch_test.c -o touch_test  
  
# 在ubuntu连接开发板的adb  
adb push ./touch_test /mnt/UDISK  
  
# 运行  
./touch_test /dev/input/event4  
*/  
int main(int argc, char *argv[]) {  
    int x = 0; //触摸点 x和 y坐标  
    int y = 0;  
  
    if (argc != 2) {  
        fprintf(stderr, "Usage: %s /dev/input/eventX\n", argv[0]);  
        exit(EXIT_FAILURE);  
    }  
  
    const char *device_path = argv[1];
```

```

int fd = open(device_path, O_RDONLY | O_NOCTTY | O_CLOEXEC);
if (fd == -1) {
    perror("Failed to open device");
    exit(EXIT_FAILURE);
}

if(fcntl(fd, F_SETFL, O_NONBLOCK) < 0)
{
    perror("Failed to fcntl device");
    exit(EXIT_FAILURE);
}

printf("Listening for events on %s...\n", device_path);

struct input_event in_ev = { 0 };
while (1) {
    while(read(fd, &in_ev, sizeof(in_ev)) > 0) {
        if(in_ev.type == EV_REL) {
            if(in_ev.code == REL_X) x += in_ev.value;
            else if(in_ev.code == REL_Y) y += in_ev.value;
        }
        else if(in_ev.type == EV_ABS) {
            if(in_ev.code == ABS_X || in_ev.code == ABS_MT_POSITION_X) x = in_ev.value;
            else if(in_ev.code == ABS_Y || in_ev.code == ABS_MT_POSITION_Y) y =
in_ev.value;
            else if(in_ev.code == ABS_MT_TRACKING_ID) {
                if(in_ev.value == -1)      printf("松开 (x:%d, y:%d)\n", x, y);
                else if(in_ev.value == 0)  printf("按下 (x:%d, y:%d)\n", x, y);
            }
        }
        else if(in_ev.type == EV_KEY) {
            if(in_ev.code == BTN_MOUSE || in_ev.code == BTN_TOUCH) {
                if(in_ev.value == 0)      printf("松开 (x:%d, y:%d)\n", x, y);
                else if(in_ev.value == 1)  printf("按下 (x:%d, y:%d)\n", x, y);
            }
        }
    }
    usleep(30000);
}

close(fd);
return 0;
}

```

编译:

```

export PATH=$PATH:/home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-
V1/prebuilt/rootfsbuilt/arm/toolchain-sunxi-glibc-gcc-830/toolchain/bin
export CROSS_COMPILE=arm-openwrt-linux-gnueabi
export STAGING_DIR=/home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-
V1/out/t113_s4/100ask_lvgl9/openwrt/staging_dir/target

arm-openwrt-linux-gnueabi-gcc touch_test.c -o touch_test

```

上传到开发板：

```
# 在ubuntu连接开发板的adb  
adb push ./touch_test /mnt/UDISK
```

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- 社区交流：<https://forums.100ask.net>
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- 微信交流群：添加微信：baiwenkeji_fae 验证备注：LVGL